VALUATIONS and ABSOLUTE GALOIS GROUPS 2

last time $G_{\mathbb{Q}_p}$ small i.e.
has finitely many open subgroups of index n $\forall n$

LEMMA 3.19

K field q any prime
if K has a small absolute Galois group and char $K \neq q$

$$\Rightarrow \dim_{\mathbb{F}_q}(K^\times / K^{\times q}) < \infty$$

proof

$L = K(\zeta_q)$ Galois extension of K of degree $d | q-1$

$N_{L/K} : L^\times \to K^\times$

$\overline{N_{L/K}} : L^\times / L^{\times q} \to K^\times / K^{\times q}$ surjective since $\forall x \in K^\times$ we have $\langle N_{L/K}(x) \rangle = \langle x^d \rangle \equiv \langle x \rangle$ mod $K^{\times q}$

$$\Rightarrow \dim_{\mathbb{F}_q}(K^\times / K^{\times q}) \leq \dim_{\mathbb{F}_q}(L^\times / L^{\times q}) < \infty$$

Kummer theory $\Leftrightarrow$ only finitely many C_q extensions (G_K small)

$\square$

$\Rightarrow$ For $\mathbb{Q}_p$ $K^\times / K^{\times p}$ is finite

FIELDS WITH FINITE P-RANK

LEMMA 3.20 (Pop's lemma)

(K,v) valued field of char $(0,p)$

assume $K^\times / K^{\times p}$ is finite

then (i) Kv is perfect

(ii) if v is of rank 1 $\Rightarrow$ vK is either discrete or p-divisible

(iii) if vK is discrete $\Rightarrow$ Kv is finite

proof

$$\mathcal{O}_v^{\times p} = \mathcal{O}_v^{\times} \cap K^{\times p}$$

$$(1+\mathcal{M}_v)^p = \mathcal{O}_v^{\times p} \cap (1+\mathcal{M}_v)$$

$$x^p \equiv 1 \bmod \mathcal{M}_v \Rightarrow x \equiv 1 \bmod \mathcal{M}_v$$

1

$$1 \to \mathcal{O}_v^\times / \mathcal{O}_v^{\times p} \to K^\times / K^{\times p} \to vK / pvK \to 1$$

$$\begin{tikzcd}
1 \arrow[r] & (\mathcal{O}_v^{\times p}) \arrow[r] \arrow[d] & K^{\times p} \arrow[r] \arrow[d] & pvK \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & \mathcal{O}_v^{\times} \arrow[r] \arrow[d] & K^{\times} \arrow[r] \arrow[d] & vK \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & \mathcal{O}_v^{\times}/(\mathcal{O}_v^{\times})^p \arrow[r] & K^\times / K^{\times p} \arrow[r] & vK/pvK \arrow[r] & 1
\end{tikzcd}$$

nine lemma

$0 \to A \to B \to C \to 0$ exact $C \cong B/A$

$$1 \to (1+\mathcal{M}_v)/(1+\mathcal{M}_v)^p \to \mathcal{O}_v^\times / \mathcal{O}_v^{\times p} \to (Kv)^\times / (Kv)^{\times p} \to 1$$

$$\left|(K_v)^\times/(K_v)^{\times p}\right| \left|(1+M_v)/(1+M_v)^p\right| = \left|\mathcal{O}_v^\times/\mathcal{O}_v^{\times p}\right| \leq \left|K^\times/K^{\times p}\right|$$

(i) K_v perfect (if K_v finite v)

$|K_v| = \infty \Rightarrow |(K_v)^p| = \infty$

let $x \in (K_v)^\times$

$|\{x + a^p \mid a \in (K_v)^\times\}| = \infty$ $\quad |K^\times/K^{\times p}| < \infty$

pigeonhole principle

$a, b, c, d \in (K_v)^\times \quad a \neq c, b \neq d$

$(x+a^p)b^p = (x+c^p)d^p \Rightarrow x = \frac{(cd-ab)^p}{(b-d)^p} \in (K_v)^{\times p}$

$\Rightarrow K_v$ perfect

(ii) $(1+M_v)/(1+M_v)^p$ finite

$\{1+a_i\}_{1\leq i\leq n} \subset 1+M_v$ complete set of representatives

$\delta = \min\{va_1, \dots, va_n, vp\} > 0$

let $x \in M_v \Rightarrow 1+x = (1+a_i)(1+y)^p$

$\Rightarrow vx = v(a_i(1+y)^p + [(1+y)^p - 1 - y^p] + y^p) \geq \min\{va_i, v_p, v(y^p)\}$

if $vx < \delta \Rightarrow v(y^p) < \delta \Rightarrow vx = v(y^p) \in p\, vK$

δ — minimal possible in $vK \Rightarrow vK = \delta\mathbb{Z}$ discrete

— otherwise $(0,\delta) \cap vK$ is p-divisible and nonempty

vK rank 1 $\Rightarrow (0,\delta)$ generates vK

$\Rightarrow vK$ p-divisible

(iii) K has an uniformiser

$(1+M_v)^p \subseteq 1+M_v^2$

in particular $\underbrace{\left|(1+M_v)/(1+M_v^2)\right|}_{=|K_v|} \leq \left|(1+M_v)/(1+M_v)^p\right| < \infty$ $\Rightarrow$ finite

GALOIS CHARACTERISATION OF HENSELIANITY

Def

(K,v) valued field p any prime

v is tamely branching at p if

(i) $\mathrm{char}\, Kv \neq p$ (residue field char $\neq p$)

(ii) $vK \neq p\,vK$ (value group not p-divisible)

(iii) $p^2 \mid |G_{Kv}|$ whenever $(vK : p\,vK) = p$

Remark

$\mathrm{char}\, k = 0$ field does not admit a non-trivial henselian valuation (e.g. $k = \mathbb{Q}$)

(K,v) henselian valued field with divisible value group vK and residue field $Kv = k$

$K = \bigcup_{n>0} k((t^{1/n}))$ field of Puiseux series over k

(valuation over t)

$\Rightarrow G_K \cong G_k$ by Fact 2.21 $\to R = \{0\}$

2.22 $\to R$ is the unique Sylow p-subgroup of I $\mathrm{char}\, k = p$

$D = \{\sigma \in G \mid \sigma \mathcal{O}_w = \mathcal{O}_w\} = \{\sigma \in G \mid \sigma(x) - x \in \mathcal{O}_w \ \forall x \in \mathcal{O}_w\}$

decomposition group

$I = \{\sigma \in G \mid \sigma(x) - x \in \mathcal{M}_w$ for all $x \in \mathcal{O}_w\} \trianglelefteq D$ and closed

inertia group

$R = \{\sigma \in G \mid \sigma(x) - x \in x\mathcal{M}_x \ \forall x \in \mathcal{O}_w\} \trianglelefteq I$

ramification group

$$\begin{array}{c} G \\ | \\ D \\ | \\ I \\ | \\ k \end{array} \qquad 1$$

$$\begin{array}{c} 1 \\ \| \\ R \\ \trianglelefteq \\ I \\ \trianglelefteq \\ D \\ | \\ G \end{array}$$

$e(w/v) = p$ power

1	L		wL
$\mathrm{Syl}_p(I)$	purely wild	purely inseparable	
R	L^R		$\frac{1}{p^\infty} vK \cap wL$
	totally tamely ramified		prime to p divisible hull
I	L'		vK
	unramified		$=$
D	L^D		vK
	immediate separable		$=$
G	K		vK

$e(w/v) = f(w/v) = 1$

$(L,w)/(K,v)$

inert $f(w/v) = [L:K]$
& Lw/Kv separable

totally ramified $e(w/v) = [L:K]$
tame $\hookrightarrow$ $p \nmid e(w/v)$, char $Kv = p$
and separable

$e(w/v) = (wL : vK)$
$f(w/v) = [Lw : Kv]$

(b) $K = \mathbb{C}((t))$ t adic valuation
$G_K \cong G_{\mathbb{F}_q}$ ← any finite field
value group $\mathbb{Z}$
residue field $\mathbb{C}$ however $R_{\mathbb{C}((t))} = 1$

$G_K = D_K \cong I_K \cong \hat{\mathbb{Z}} = \prod_q \mathbb{Z}_q$

|| we can't expect to recover the t-adic valuation from the isomorphism class of G_K

if $\mathrm{char}(Kv) \neq p$ $(vK : p\,vK) = p$ and $p \nmid G_{Kv}$

$\Rightarrow G_K$ has Sylow p-subgroup $\mathbb{Z}_p$

$G_K = \mathrm{Gal}($

THEOREM 3.29

A field K admits a henselian valuation tamely branching at p $\Leftrightarrow$ G_K has a Sylow p-subgroup $P \not\cong C_2, \mathbb{Z}_p, \mathbb{Z}_2 \rtimes C_2$ that contains a non-trivial normal abelian closed subgroup N.

LEMMA 3.32

K field with henselian v tamely branching at p
w its unique extension to $L = \mathrm{Fix}\, P$ P Sylow p-subgp. of G_K

$\Rightarrow 1 \neq \underset{\substack{\text{abelian} \\ \text{closed}}}{I_L} \trianglelefteq P$ and $P \not\cong C_2, \mathbb{Z}_p, \mathbb{Z}_2 \rtimes C_2$

<u>proof</u> fact 2.21

$D_L = G_L = P$

$I_L \underset{\text{closed}}{\trianglelefteq} P$

fact 2.22
$R_L = 1$ since $\mathrm{char}\, Kv \neq p$

<u>THM 2.27</u> $L = K^{sep}$

$$D_K / R_K \cong \mathrm{Hom}(wL/vK, \mu(Lw)) \rtimes G_{Kv}$$

$$\cong \prod_{q \neq p} \mathbb{Z}_q^{r_q}$$

$r_q = \dim_{\mathbb{F}_q} (vK / qvK)$